

William C. Hernandez

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EDUCATION

University of Waikato

Hamilton, Waikato

Bachelor of Software Engineering with Honours

Expected Graduation, Nov 2028

- **Focus Areas:** Machine Learning, Embedded Systems, Full-Stack Development.
- **Extracurriculars:** Young Engineers Society, Waikato Astronautics Club (UWAC), Class Representative for 10 Papers, CS³ (Computer Science Club), MC² (Mathematics Club).

St. John's College

Hamilton, Waikato

NCEA Level 1, 2 and 3 Excellence Endorsement

Jan 2020 - Nov 2024

- **Leadership Role:** Arts Leader, Student Prefect, Student Volunteering Leader.

WORK EXPERIENCE

AI Trainer

DataAnnotation (Freelance/Remote)

Apr 2026 - Jun 2026

- Designing complex tasks to train and identify vulnerabilities and weaknesses in LLMs.

SCMS Sessional Assistant

University of Waikato (Contract/On Site)

Mar 2026 - Present

- Facilitated technical labs for 50+ students learning to program in Python, MATLAB, C#, Verilog and Assembly (WRAMP) across 100 and 200 level papers. ([COMPX230](#), [COMPX161](#), [COMPX101](#), [ENGEN103](#)).
- Conducted live code reviews, evaluating code quality integrity to provide actionable, real-time feedback to students.
- Marked and provided feedback to a range of practical (coding) and written tests.

Engineering Student Ambassador

University of Waikato (Contract/Outreach)

May 2025 - Present

- Representing the University of Waikato in Outreach programmes across the North Island related to Software Engineering.

ACTIVITIES & LEADERSHIP

University of Waikato Astronautics Club (UWAC)

General Executive (IT Admin), Marketing Assistant

2026 - Present

- **Firmware Engineering:** Developing and optimising flight computer firmware using C++ and Arduino to process high-frequency data and real-time telemetry. Gained familiarity with KiCAD and embedded systems to ensure low-latency hardware-software communication.
- **Information Security & IT Admin:** Managing ITAR-compliant data environments (SharePoint Infrastructure), ensuring strict access control and secure versioning for sensitive information.
- **Full-Stack Engineering:** Developing a comprehensive redesign of the [uwac.nz](#) website using React and Cloudflare D1 and R2 databases.. Implementing a relational database to handle member data and real-time activity.

TECHNICAL PROJECTS

Anatomy Explorer

Co-Developer · COMPX241 - Smoke & Mirrors

2026

- Created a 3D anatomical web application as a group using a React/Three.js framework.
- Developed dynamic scaling systems using OLS regression, and multiple interactive layers.
- Implemented a dynamic quiz system with multiple modes to target specific body regions.

UWAC Beginner's Rocketry Firmware 2026

Firmware Developer · University of Waikato Astronautics Club

2026

- Developed the firmware for the beginners rocketry programme using C++.
- Designed to safely decouple at apogee and deploy a main parachute at 300m AGL
- Features include IMU-based flight state detection and LED blink codes for diagnostics.

CampusBite - Find extra food on campus

Full-Stack Developer · UNIHACK AU/NZ Hackathon

2026

- Real-time peer-to-peer food networking application built on React using Firebase for live activity.
- Uses Gemini Vision API to recognise photos of food and precision matches to user dietary requirements. Features a secure claim system with pickup verification codes and user histories showcasing proficiency in handling in live, synchronised data streams.

NurseNotes - Turning complex medical jargon into plain language

2026

Full-Stack Developer · AI Hackathon NZ 2026

- Engineered system prompts for Gemma-3n-e4b hosted locally via LM Studio to translate complex medical documentation into accessible, plain language.
- Developed a React web portal for healthcare providers to input medical records, query the local LLM endpoint API, and persist simplified outputs in JSON structure.
- Built a cross-platform mobile app using Expo (React Native) as a patient companion app, enabling real-time viewing of translated medical summaries.

HONOURS & AWARDS

Computer Science Undergraduate Scholarship

University of Waikato

2024

- Awarded to the top 10 students nationwide based on a rigorous CS examination/competition.

Ko te Tangata Scholarship

University of Waikato

2024

- Prestigious school-leaver award recognising top-achieving academic students entering the university.

David Johnstone Charitable Trust Scholarship

Perpetual Guardian

2024

- Awarded to 20+ young people for demonstrating exceptional initiative and leadership.

PROGRAMMES

ELEVATE Extension Programme

Virscient

Jul 2026 - Oct 2026

- One of ten selected students to participate in an extension programme with a focus on embedded systems and radio tools.

SKILLS

- **Languages:** Python, C, C++, C#, Java, JavaScript, TypeScript, MATLAB, SQL, Verilog
- **Embedded & Hardware:** STM32, Arduino, Raspberry Pi, FreeRTOS-style task scheduling, I2C/SPI/UART, GPIO and interrupt handling, barometric/IMU/GPS sensor integration RTL
- **Web & Cloud:** React, Next.js, Vite, Node.js, Tailwind CSS, shadcn/ui, Supabase (PostgreSQL), Firebase, Cloudflare Pages/D1/R2, Vercel, Stripe, Entra ID authentication
- **Tools & Platforms:** Git, Linux, Docker, CI/CD (GitHub Actions), SharePoint, Power Apps

REFERENCES ON REQUEST